CHEMISTRY ADAPTLY

ATOMIC STRUCTURE – ATOMS, IONS & ISOTOPES

KEY VOCABULARY

Atomic number: the number of protons in an atom.

Mass number: the total number of protons and neutrons.

Isotope: atoms of the same element with different numbers of neutrons.

Ion: a charged atom formed by losing or gaining electrons.

COMMON MISCONCEPTIONS

~~Atoms always have equal protons and neutrons.~~

✓ Neutron number often differs from proton number.

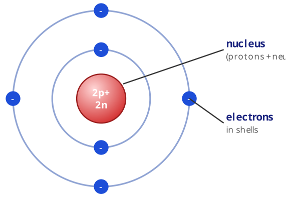
~~Isotopes have different chemical reactions.~~

✓ Isotopes react the same — they share electron arrangement.

~~The mass number is the mass in grams.~~

✓ It is just the count of protons + neutrons.

STRUCTURE OF THE ATOM



A tiny **nucleus** of protons and neutrons is surrounded by **electrons** in shells.

1 RELATIVE MASS & CHARGE

State the **relative charge** of a proton, a neutron and an electron. **[3 marks]**

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2 READ THE SYMBOL

A sodium atom is written as $^{23}{}_{11}Na$. Give the number of **protons, neutrons and electrons**. **[3 marks]**

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3 ELECTRONIC STRUCTURE

An atom has 17 electrons. Write its **electronic structure**. **[1 mark]**

Fill the inner shells first (2, 8, 8 ...).

Answer: ______________________________ (e.g. 2,8,...)

4 ISOTOPES

Chlorine has isotopes ^{35}Cl and ^{37}Cl. **Explain** why they are isotopes of the same element. **[2 marks]**

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5 FORMING IONS

Complete the sentence using the word bank. **[2 marks]**

When an atom _____ electrons it becomes a _____ ion.

loses | positive | gains | negative

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6 RELATIVE ATOMIC MASS

Chlorine is 75% ^{35}Cl and 25% ^{37}Cl. **Calculate** its relative atomic mass. **[2 marks]**

Answer: ______________________________ (A_r)

TIP: *Protons + neutrons sit in the nucleus; electrons orbit in shells.*

✓ Think: Why do isotopes of an element have identical chemical properties?